Subject-independent forearm sEMG electrode reduction with cross-fitted attribution and non-inferiority testing

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Abstract

Objective. Determine how many forearm surface electromyography electrodes can be removed while preserving mean gesture recognition for unseen participants, and assess the additional value of attribution-based selection. *Approach.* We analyzed 43 GRABMyo participants using sixteen monopolar electrodes and sixteen active gestures. Participant-disjoint inner cross-fitting estimated TreeSHAP rankings within each outer leave-one-subject-out fold. Reduced models were retrained and compared with full16 using participant-bootstrap intervals and a predefined 0.02 Macro-F1 engineering tolerance. Exploratory checks included primary-budget geometric selection, paired gesture predictions, shared-resource permutation selection, and task, window and classifier sensitivities. *Main results.* SHAP12 achieved mean Macro-F1 0.528 versus 0.541 for full16; its change was -0.0133 (95% CI -0.0160 to -0.0106). Twelve was the smallest qualifying count. Nevertheless, 11 participants lost more than 0.02 and 4 gestures showed corrected declines. Geometric12 did not establish non-inferiority, and the direct SHAP12 mean advantage remained uncertain (+0.0060; 95% CI -0.0010 to +0.0127). Attribution outperformed mean random subsets in the lighter comparison. Shared-resource SHAP outperformed grouped permutation at six electrodes, but its twelve-electrode advantage remained uncertain. Stability regularization did not establish an improvement. *Significance.* One quarter of this monopolar array could be removed within the mean tolerance under offline participant-independent evaluation. This validates a selection procedure rather than one locked physical layout; individual losses and uncertain transfer require independent validation before device claims.

1 Introduction

Surface electromyography (sEMG) provides a non-invasive signal for gesture recognition and myoelectric control, but the recording interface must balance spatial information against the burden of additional electrodes (Scheme and Englehart, 2011; Farina et al., 2014; Vujaklija et al., 2016). Fewer recording sites could simplify an interface. Whether that reduction preserves recognition for an unseen participant is an empirical question, and offline recognition alone does not establish a usable control interface.

Channel selection has been studied with feature filters, wrappers and spatial criteria (Phinyomark et al., 2012; Stango et al., 2015; Qu et al., 2021). SHAP-guided EMG channel and feature optimization is also established (Rani et al., 2025); related work uses Shapley explanations for similar sEMG gestures (Wang et al., 2024) and explainable feature selection with fusion (Gehlot et al., 2024). These contributions make attribution-based reduction a relevant comparator rather than a new algorithmic premise. Their performance estimates cannot be directly compared with this study across different tasks, sensors and validation protocols.

For a system intended to work across people, selection must exclude the eventual test participant. Using that participant to standardize features, estimate importance or choose a subset can produce optimistic estimates. Participant-disjoint inner evaluation also avoids measuring attribution solely on observations used to fit the attributing model. This motivates an outer leave-one-subject-out (LOSO) evaluation in which each entire selection procedure is reconstructed from the outer training participants.

Attribution magnitude describes predictive reliance of a fitted model, not a direct measurement of muscle activation. TreeSHAP distributes a model's prediction over its input features (Lundberg and Lee, 2017; Lundberg et al., 2020); aggregation over an electrode's features offers a ranking, but does not turn it into a coalition-level Shapley value for that electrode. Correlated features and variation across participants can affect this ranking. Attribution stability has already motivated resampling and regularization methods (Visani et al., 2022; Xiang et al., 2026); a stability penalty therefore requires validation rather than a novelty claim.

Electrode count requires an explicit tolerance for performance loss. Non-inferiority asks whether a reduced system remains within such a tolerance, whereas a non-significant superiority test does not establish preservation (Schuirmann, 1987; Walker and Nowacki, 2011). A mean tolerance is also different from preserving every participant or gesture. Reporting only an aggregate curve could hide losses that matter to an intended application.

We evaluate cross-fitted electrode reduction on GRABMyo and separately examine NinaPro DB5. The primary question is the minimum count supported by an internally documented, predefined engineering margin. Secondary questions concern utility beyond random and spatial selection, stability regularization, and deletion with retraining. New exploratory checks use a geometric subset under the primary training protocol, paired participant and gesture predictions, and shared-resource SHAP and grouped permutation rankings. Sensitivities and split-derived layouts delimit the claim: the LOSO experiment evaluates a selection procedure, not a prospectively validated common physical layout.

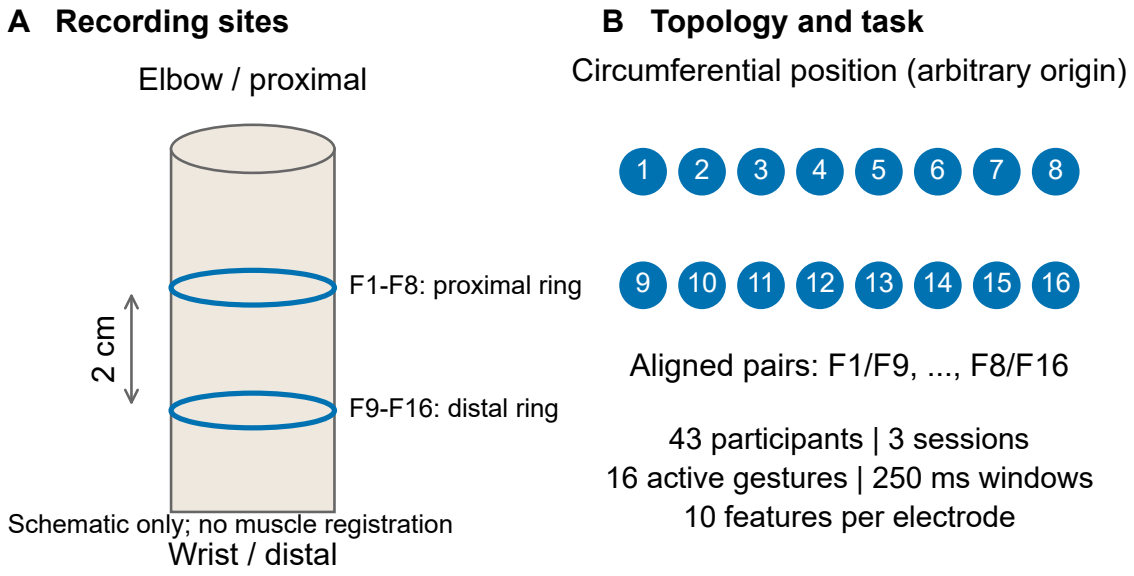


Figure 1. Recording topology and task. The source montage places F1–F8 in the proximal ring and F9–F16 in the distal ring, separated by 2 cm. Corresponding circumferential positions are aligned. The forearm outline is schematic; recording sites were not registered to individual muscles. The unwrapped map preserves electrode numbers without implying a verified absolute anatomical angle.

2 Methods

2.1 Datasets, selection units and signal processing

GRABMyo v1.1.0 contains 43 able-bodied participants recorded over 3 days with an EMGUSB2+ amplifier at 2048 Hz (Pradhan et al., 2022; Jiang et al., 2024). Source-reported demographics were 23 male and 20 female, age 26.4 ± 2.9 years. The forearm montage has sixteen monopolar sites in two rings (figure 1); each site supplies one channel, so electrode and channel selection coincide here. Counts refer only to these signal recording sites; reference and ground requirements are not evaluated. The primary task uses sixteen active gestures, excluding rest. All 30,715 distributed files matched published checksums, and 15,351 records were processed. Source demographics and the official gesture dictionary are provided with the reproducibility package.

Signals were filtered at 20–450 Hz using a Butterworth band-pass with SciPy order argument 4 (eight poles per pass), applied forward and backward for zero phase, followed by a forward–backward 50 Hz notch ($Q = 30$). They were then divided into 250 ms windows with 50% overlap within trial boundaries. The non-causal filter limits the analysis to offline use. Each channel supplied mean absolute value, RMS, variance, waveform length, zero crossings, slope sign changes, integrated EMG,

Willison amplitude, mean frequency and median frequency. The active-task cache contains 563,472 windows with 160 features each.

Count-feature thresholds followed a common rule: 1.15 times the median absolute signal first difference, rounded to one significant figure. Historically, the monopolar threshold (10^{-2} mV) was derived from 54 unfiltered records from six participants before LOSO. It was not estimated separately within each outer fold. Subsequent exclusion checks reproduced the identical rounded threshold in all 43 folds. Thus the stored features are invariant to this particular exclusion, but the original development sequence should not be described as fully isolated. Fold-specific feature means and standard deviations, classifier fitting and attribution ranking exclude the outer test participant.

NinaPro DB5 was analyzed separately: 10 participants, two Myo armbands, sixteen channels sampled at 200 Hz and exercise 1's twelve finger movements (Pizzolato et al., 2017). Refined labels determine movement segments. Filtering was effectively 10–89 Hz without a notch; the requested 95 Hz upper limit was clipped by the shared implementation to $0.45f_s - 1$ Hz. Complete raw reconstruction of all saved DB5 feature rows verified this implemented cutoff; the earlier 95 Hz description was a reporting error. The 250 ms windows contain 50 samples. The threshold is seven quantization steps in the stored integer representation; every participant-exclusion check reproduced it. Source-reported DB5 demographics were 8 male and 2 female, age 28.0 ± 4.0 years. Sex is reported as recorded in the source datasets; no sex-specific effect is estimated. No anatomical correspondence between DB5 and GRABMyo channel numbers is assumed.

Ethics This is a secondary analysis of publicly distributed, de-identified recordings, with no new recruitment or recording. Original study approvals and consent are described in the dataset publications. [AUTHOR INPUT REQUIRED: institutional determination or exemption for this secondary analysis remains unresolved.]

2.2 Outer evaluation and classifier budgets

Each outer LOSO fold reserves all observations of one participant p. For cross-fitted attribution, an inner five-fold GroupKFold partitions only the other participants (figure 2). Inner feature standardization uses inner-training observations only; final reduced models use the corresponding outer-training statistics. No outer test observations enter fitting or rankings.

The primary reduction curve uses XGBoost with 300 boosting rounds, maximum depth 6, learning rate 0.1, row and column subsampling 0.8, histogram tree construction and fixed seed 20260728. Its multiclass soft-probability objective produces probabilities whose argmax gives the predicted class. For the primary curve, final models across all counts and fixed penalty rankings within an outer fold use the same stratified sample of 200,000 outer-training windows. Models are retrained from scratch on the selected features.

Training budgets differ across historical tracks: the full-array XGBoost classifier context uses all training windows, the primary curve uses 200,000, and the random-selection comparison uses 50 boosting rounds and 150,000. The full-array SVM uses 40,000 windows; random forest uses 200,000. These are budget-limited comparators, not proven lower bounds or matched classifier-capacity experiments. Budgets were calibrated using two outer-fold outcomes. The internally dated analysis plan followed an earlier pilot; no independent registration or Git commit authenticating its date is available. These development exposures qualify claims of wholly prospective evaluation.

2.3 Cross-fitted attribution and its participant-level extension

Inner XGBoost models use at most 200,000 training windows sampled uniformly without replacement after participant splitting. Their standardization statistics use the full inner-training pool, not the capped model-fit sample. Exact tree-path-dependent TreeSHAP contributions are computed on raw class margins for at most 20,000 inner-held-out windows. For electrode c, absolute contributions are averaged across windows, classes g and its ten features f:

$$\Phi_{i,c} = \frac{1}{GF} \sum_{g=1}^{G} \sum_{f=1}^{F} |\phi_{i,g,c,f}|. \tag{1}$$

The window mean is normalized across electrodes to give the inner-fold score $I_{j,c}$. The central magnitude and unscaled median absolute deviation are

$$A_c = \text{median}_j\, I_{j,c}, \quad U_c = \text{median}_j\, |I_{j,c} - A_c|, \quad Q_c(\lambda) = A_c - \lambda U_c. \tag{2}$$

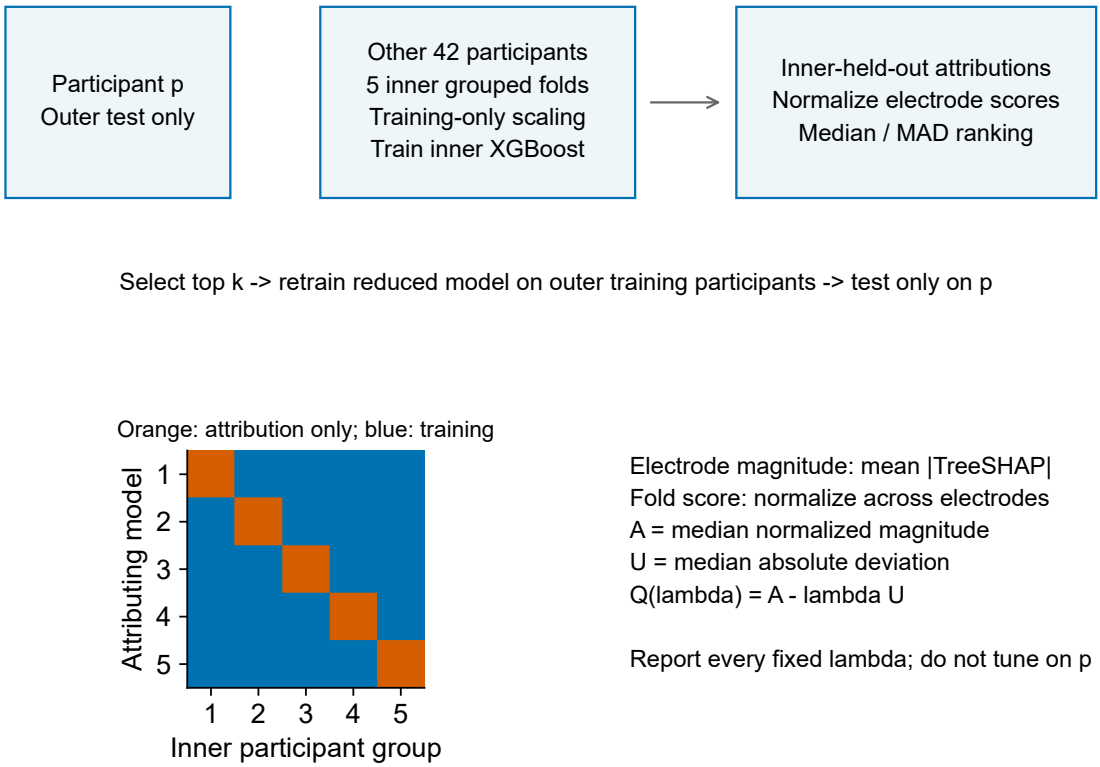


Figure 2. Participant access in the nested procedure. Each inner attributing model excludes the participants on whom its attributions are evaluated. The outer participant is reserved until final reduced-model testing. This diagram describes fold-specific scaling, fitting and ranking; historical threshold pooling and budget calibration are separate development exposures documented in the text.

All fixed penalties $\lambda \in \{0, 0.25, 0.5, 1\}$ are reported, with lower electrode index breaking ties. No penalty is chosen using outer scores. The unpenalized ranking is called cross-fitted SHAP; the positive penalties assess stability regularization.

Pooling people within five inner folds can suppress person-level heterogeneity. A post-hoc extension therefore computes the median and MAD over normalized individual attribution vectors of all 42 outer-training participants. These vectors share five fitted inner models; their dispersion is descriptive heterogeneity, not a standard error. All penalties and counts are evaluated under the unchanged primary protocol. Exact matching participant/subset fits may be reused with source-row provenance; new subsets require new fits. A historical source-hash mismatch is preserved in the audit, and current reporting uses an isolated refit with current input fingerprints. After observing the penalty results, a descriptive audit compared selected physical site sets and Jaccard similarity at every fixed penalty and count. It added no model fits or inferential tests and did not choose a penalty.

2.4 Primary endpoint and participant-level inference

Macro-F1 is the unweighted mean of the gesture F1 scores for each held-out participant. For count k, the paired change is $\Delta_{p,k} = \mathrm{F1}_{p,k} - \mathrm{F1}_{p,16}$. The predefined primary engineering tolerance is $\delta = 0.02$ on the 0–1 scale. The historical decision rule requires the lower 95% percentile-bootstrap bound on the participant mean change to exceed $-\delta$; k^* is the smallest qualifying count below sixteen. Sensitivity margins are 0.01 and 0.03. This criterion concerns mean recognition performance, has no established clinical interpretation and does not imply equivalence or individual preservation.

The participant is the statistical unit: 43 for GRABMyo, ten for DB5 and 21 or 22 test participants for split-layout checks. Bootstrap intervals use 10,000 resamples of paired participant scores with seed 20260728. They are conditional on the observed fitted LOSO models. Overlapping training cohorts and the lack of refitting within every bootstrap mean these intervals do not capture all new-cohort training uncertainty (Bengio and Grandvalet, 2004).

The original Holm procedure used the fraction of unshifted bootstrap mean differences at or below $-\delta$, with a Monte Carlo floor. This is a percentile-tail quantity rather than a bootstrap distribution generated under the null. We preserve it and additionally report an exploratory, null-

centred studentized bootstrap (Davison and Hinkley, 1997). For $d_p = \Delta_{p,k}$, residuals $d_p - \bar{d}$ are resampled, their studentized means are compared with $(\bar{d}+\delta)/(s_d/\sqrt{n})$, and the upper-tail probability uses a plus-one correction. Holm adjustment covers the fifteen reduced-count tests at one-sided $\alpha = 0.025$, separately for each condition, margin and penalty. No seed or margin is chosen from results.

Paired secondary contrasts use Wilcoxon signed-rank tests and participant mean-change percentile intervals. These address different estimands: signed ranks are not a test of the mean. Holm families are stated explicitly; a non-significant contrast is not evidence of equivalence. The detailed historical secondary families were specified after their underlying results became available and are labeled secondary rather than prospectively confirmatory.

2.5 Selection comparators and attribution diagnostics

Historical lighter-model comparisons use random subsets, cross-fitted SHAP, stability-regularized SHAP ($\lambda = 0.5$), feature-wise permutation, mutual information and geometric farthest-point selection. Random selection uses 100 subsets per participant at counts 4, 6 and 8, and 500 at count 12. Eight attribution-versus-mean-random contrasts receive Holm correction. Historical permutation uses one model fitted on 20,000 windows and three feature-wise permutations on 8,000 non-overlapping windows from the same outer-training participant pool. Its 40,000-window ranking pool was standardized before the window split; this was not participant-disjoint inner validation. The outer test participant remained excluded. SHAP instead uses the participant-disjoint cross-fitting above. Thus shared final evaluation does not equalize their selection resources or validation designs.

New exploratory geometry12 testing uses the primary 300-round, 200,000-window protocol, the same outer training sample and full16 reference. Geometric selection uses the fixed two-ring topology, nominal 24 cm circumference and 2 cm separation, starts at F1 and successively maximizes distance from selected sites; ties favor the lower index. It is not tuned to participant outcomes.

A separate exploratory resource-matched comparison shares five inner models between SHAP and grouped-electrode permutation, with up to 200,000 uniformly sampled training and 2,000 validation windows per fold. Equal validation quotas represent every inner-held-out participant. All ten features of one electrode are permuted jointly within each participant, three repeats, scoring the equally weighted mean participant Macro-F1 loss. Signed losses are retained, normalized by their summed absolute magnitude, and aggregated by the median across inner folds. SHAP uses the identical observations and normalizes absolute contribution magnitudes. Final models at counts 6 and 12 use the primary protocol; two contrasts receive Holm correction. Measured training and selection times are reported separately. CPU RF fits overlapped with early folds of this experiment, so these are observed workstation timings rather than an isolated runtime benchmark. Because permutation grouping, participant-disjoint validation, scaler isolation and resources all differ from the historical method, this experiment does not isolate a budget effect.

Deletion with retraining uses the unpenalized primary ordering and compares removal of high-ranked, random and low-ranked electrodes using the lighter model. Its fifteen contrasts form one Holm family. This diagnostic tests predictive ranking relevance; it cannot identify causal muscles or validate an online controller.

2.6 Participant and gesture degradation

Primary full16, SHAP12 and geometry12 models were rerun to retain predictions indexed to the same canonical outer-test rows. Full16 and SHAP12 scores must reproduce the original curve within 10^{-6}. Participant losses, quantiles, worst loss and the number exceeding the primary margin are descriptive. Gesture-specific F1 differences are paired over all participants, with Holm correction across sixteen gestures. Confusion changes average row-normalized confusion matrices within participants so people, rather than windows, receive equal weight. Gesture names follow the explicit numeric ordering in MotionSequence.txt.

2.7 Sensitivity, transfer and fixed-layout scope

Rest-inclusive recognition and 150/500 ms windows use raw-signal re-extraction and complete count curves with the primary 250 ms ranking held fixed. The 150 ms window is 307 samples with stride 153; 250/500 ms use 512/1024 samples and half-window strides. The same 200,000-window count does not equate recording duration across windows. A random-forest sensitivity uses the pre-existing 200-tree, minimum-leaf-size-2 configuration at counts 12 and 16, the unchanged XGBoost ranking and the same training sample. It tests those counts only and cannot establish a random-forest minimum count.

DB5 applies the procedure separately with its own sixteen-channel reference. Derived bipolar signals $B_j = F_j - F_{j+8}$ use eight aligned pairs, re-extracted features and the eight-pair reference; they are not physical bipolar-sensor measurements. Full-array cross-session analyses compare training on session 1 with testing on sessions 1–3, and training on sessions 1–2 with testing on session 3. They do not test a reduced layout under displacement, and the latter comparison confounds data quantity with session diversity.

Fixed-layout checks use odd-ID development/even-ID testing and the reverse, each deriving a layout using development participants only and the original five-fold median/MAD aggregation rule. Their inner and final models use all development windows with training-only standardization and the same 300-round classifier, rather than the primary curve's 200,000-window cap. The historical pooled feature thresholds also apply to these checks. The count of twelve had already been selected using the full cohort, so neither direction independently validates count selection. The layouts differ and must be reported separately. To quantify Monte Carlo decision sensitivity near the boundary, a post-hoc diagnostic repeated the original percentile bootstrap with 200 seeds drawn once from a generator initialized at 20260728, using 10,000 resamples for each seed. Every seed outcome is retained; the diagnostic never replaces the original seed or decision. No fresh, independently collected cohort with this montage was available. Feature redundancy and ranking-agreement diagnostics are documented in the Supplementary and machine-readable outputs; high outer-fold agreement is qualified by extensive training-set overlap.

3 Results

3.1 How many electrodes meet the mean performance tolerance?

Under the primary curve protocol, full16 achieved mean Macro-F1 0.541 and SHAP12 achieved 0.528. Twelve electrodes was the smallest count meeting the 0.02 margin, with paired change -0.0133 (95% CI -0.0160 to -0.0106). Eleven did not meet this criterion: change -0.0204 (-0.0246 to -0.0162). The original percentile decision and the exploratory null-centred, Holm-adjusted sensitivity gave the same primary minimum count: 14, 12 and 11 at margins 0.01, 0.02 and 0.03, respectively. The original four penalty curves did not change these count decisions (figure 3; full results in the Supplementary).

Mean preservation did not extend uniformly to individuals. For SHAP12 minus full16, the median change was -0.0125 and participant SD was +0.0093. 1 participant improved, while 11 lost more than 0.02. The worst observed change was -0.0361 (participant 27), and the best was +0.0144. The margin is a mean-level engineering criterion, so these counts do not redefine individual non-inferiority.

Table 1. Matched primary protocol: reduced subset minus full16. Each row uses all 43 paired participants, 300 boosting rounds and 200,000 training windows.

Subset	Mean F1	$\bar{\Delta}$	95% CI	NI at 0.02
shap12	0.528	-0.0133	[-0.0160, -0.0106]	yes
geom12	0.522	-0.0193	[-0.0254, -0.0132]	no

3.2 What does attribution add beyond random and spatial selection?

In the original lighter-model experiment, attribution-ranked subsets outperformed the participant's mean random-subset baseline at all four tested counts after correction. For the stability-regularized ranking, the mean advantages at counts 4, 6, 8 and 12 were +0.0172, +0.0370, +0.0243 and +0.0074. These absolute scores belong to the lighter comparison model and should not be substituted for the primary non-inferiority curve.

With the primary model and training budget matched, geometric12 scored 0.522, with change -0.0193 (95% CI -0.0254 to -0.0132) from full16. Its lower bound crossed the 0.02 tolerance. The direct SHAP12-minus-geometric12 mean contrast was +0.0060 (-0.0010 to +0.0127), whose interval included zero. The nominal signed-rank $p = 0.024$ addresses ranks rather than the mean; it does not establish a positive mean advantage. Different non-inferiority decisions likewise do not establish a difference between methods (Gelman and Stern, 2006).

The shared-model comparison is reported separately in table 2 and figure 4(C). At counts 6 and 12, SHAP-minus-grouped-permutation mean changes were +0.0309 (+0.0211 to +0.0410) and +0.0002 (-0.0029 to +0.0034), with Holm $p < 0.001$ and $p = 0.976$. Mean measured per-outer-fold times were 117.1 s for shared inner training, 4.0 s for SHAP selection and 10.6 s for grouped permutation. These hardware-dependent measurements describe the executed comparison rather than online latency.

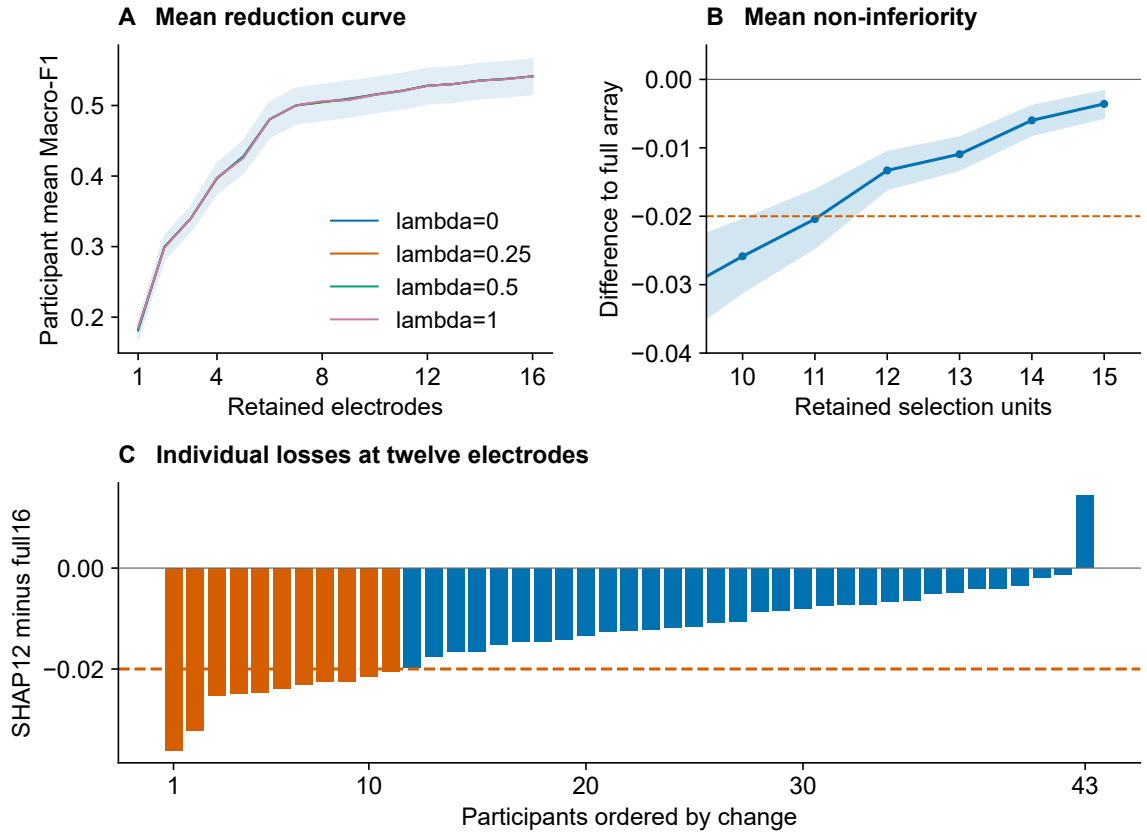


Figure 3. Count reduction and its limits. (A) Participant mean curves for all fixed penalties; shading shows the 95% participant-bootstrap interval for the unpenalized curve. (B) Mean paired change against the same curve's full16 reference, with 95% intervals and the 0.02 tolerance. (C) All participant changes at SHAP12, ordered by change; orange identifies observed losses larger than 0.02. The horizontal tolerance in C is a descriptive reference, not an individual inference threshold.

At $k = 6$, the mean interval supports a positive SHAP advantage; the corrected signed-rank contrast is significant. At $k = 12$, the mean advantage remains uncertain because its interval crosses zero; the corrected signed-rank contrast is not statistically detectable.

Table 2. Exploratory SHAP minus grouped-electrode permutation. The two methods share the same five inner models and validation observations per outer fold. Holm correction covers the two contrasts.

k	SHAP	Permutation	$\bar{\Delta}$	95% CI	p_{Holm}
6	0.480	0.449	+0.0309	[+0.0211, +0.0410]	< 0.001
12	0.528	0.528	+0.0002	[-0.0029, +0.0034]	0.976

3.3 Which predictive information is lost, and does regularization help?

Attribution has consistent spatial preferences but does not register sites to individual muscles (figure 5(A)). The same top-four set (4, 6, 7, 14) occurred in 36 outer folds. Outer-fold agreement is descriptive because training cohorts overlap heavily.

The gesture analysis found 4 corrected losses among sixteen active gestures. The largest mean decline was gesture 14, forearm pronation: -0.0456 (95% CI -0.0594 to -0.0323). Gestures 2 (thumb adduction), 9 (index finger extension) and 10 (thumb finger extension) also showed corrected declines. These paired gesture-F1 results expose task-specific losses hidden by the global mean; no anatomical cause is inferred from them. Gesture IDs are distinct from electrode indices.

Deletion with retraining produced the expected ordering for the common random-comparison counts: removing highly ranked electrodes gave lower performance than random removal, which gave lower performance than removal of weakly ranked electrodes. At six electrodes retained, Remove-Top minus Remove-Bottom was -0.0775 (-0.0921 to -0.0627; Holm $p < 0.001$). This supports predictive relevance under retraining, rather than a causal physiological interpretation.

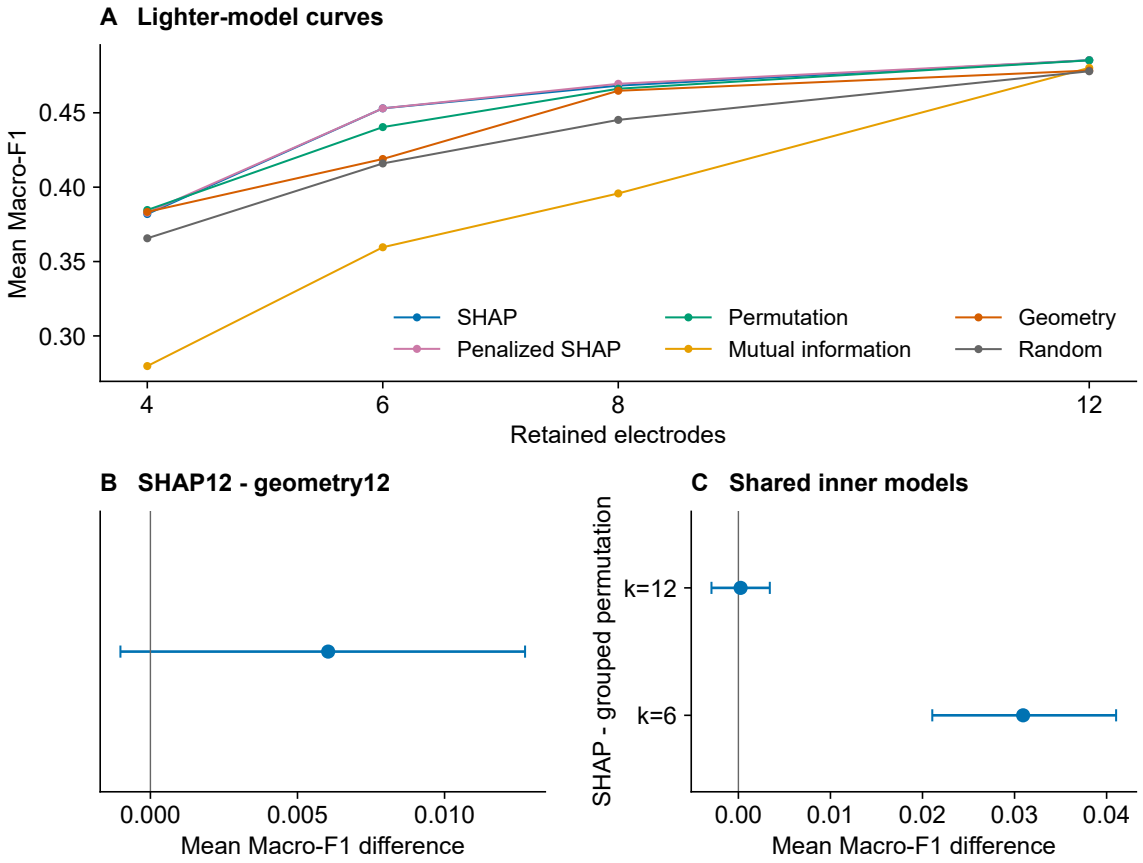


Figure 4. Incremental selection utility under distinct protocols. (A) Historical six-method comparison with a common lighter final classifier (50 boosting rounds, 150,000 training windows); selection-stage resources and validation designs differed, and penalized SHAP uses $\lambda = 0.5$. (B) Direct SHAP12-minus-geometric12 mean contrast under the primary protocol (300 boosting rounds, 200,000 windows). (C) SHAP-minus-grouped-permutation contrasts with shared inner models and validation observations and primary-protocol final models. Error bars in B–C are participant mean-change bootstrap intervals. Panels are not a common absolute-score benchmark.

None of the original 12 penalty contrasts showed corrected improvement over magnitude alone. The participant-level extension, which exposes individual heterogeneity instead of pooling it into five groups, yielded 0 corrected beneficial and 0 corrected harmful penalty contrasts across the twelve fixed comparisons (the Supplementary penalty table). All penalties remain reported; outer-test performance was not used to choose one.

At twelve electrodes, every fixed participant-level penalty selected the same physical site set as its unpenalized counterpart in all 43 outer folds. The same fixed-protocol cached fits were reused for these identical inputs, giving identical recorded scores; this is input identity rather than an equivalence test of different layouts.

3.4 Where does the electrode-count conclusion stop generalizing?

The two split-derived fixed layouts gave different results. Odd-ID development/even-ID testing removed 3, 9, 11, 12, with change -0.0124 (-0.0179 to -0.0077), meeting the 0.02 criterion. The reverse removed 9, 11, 12, 13, with change -0.0135 (-0.020020 to -0.0066), whose lower bound missed the criterion. The reverse split met the percentile criterion in 90 of 200 bootstrap seed reruns (45%); this indicates numerical decision fragility, not a probability that NI is true. The original seed decision is retained. These are two layouts rather than one layout validated twice; the count had already been chosen using the full cohort. Neither is an independent prospective validation.

Rest-inclusive, window-length and fixed-count random-forest sensitivities compare reduced systems with their own full arrays (table 3; figure 6(B)). Complete window and rest curves appear in the Supplementary. They reuse the primary ranking and test robustness of reduced-model evaluation; they do not show that newly estimated rankings would be invariant to those changes. The random-forest check has no estimated minimum count because only twelve and sixteen were tested.

DB5 provided a small, separate-data test. Its count-12 lower bound under the historical percentile criterion was -0.019994, close to the 0.02 boundary, and no count met the historical Holm rule at that

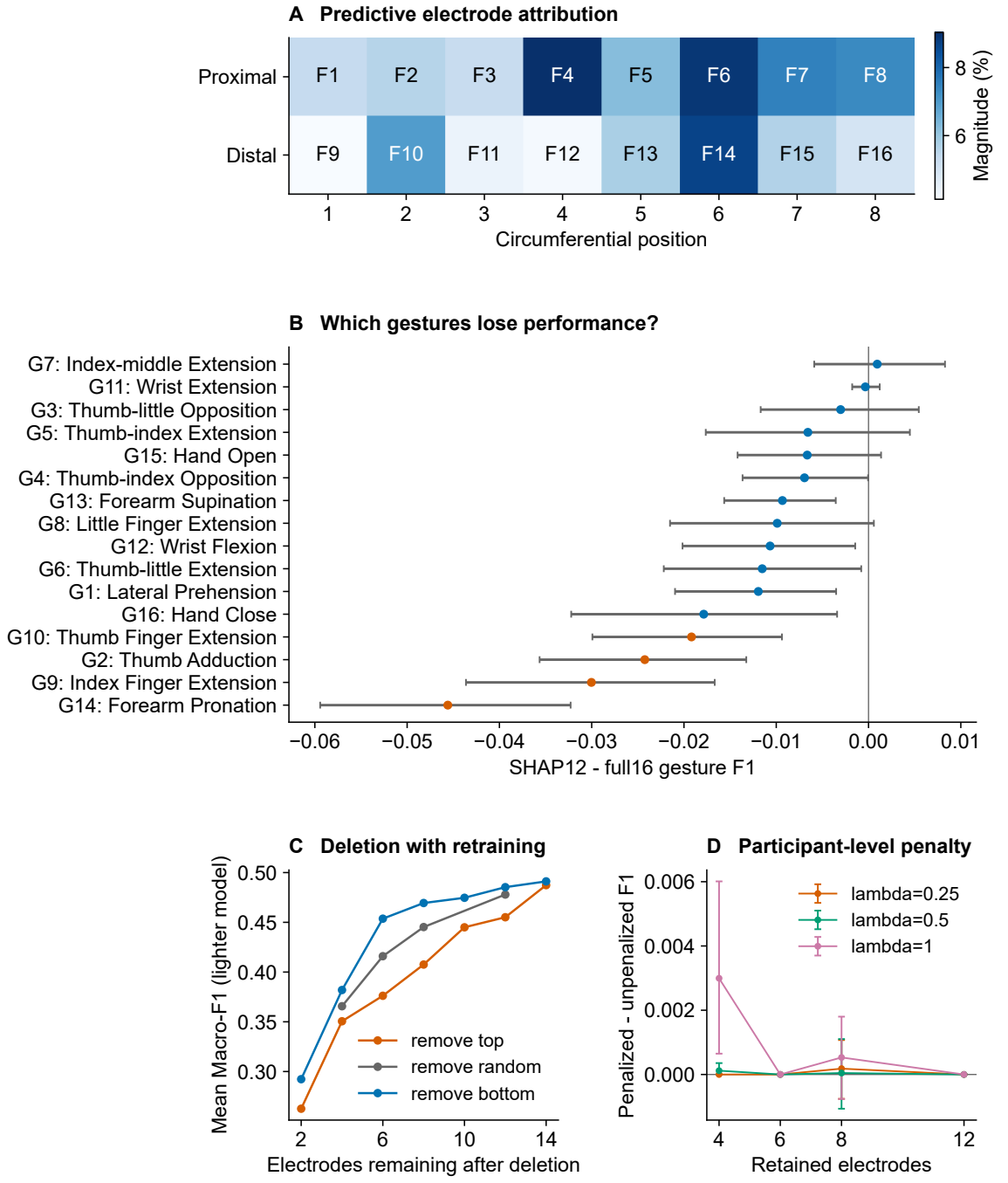


Figure 5. What the ranking represents and what reduction loses. (A) Mean fold-median normalized absolute attribution over outer folds on the documented topology; the scale is in percent and measures model relevance, not muscle activation. (B) Gesture-specific mean F1 changes with participant-bootstrap intervals; orange marks Holm-adjusted signed-rank $p < 0.05$ across sixteen gestures. Names follow MotionSequence.txt, with paired-finger names shortened for display. (C) Deletion with retraining under the lighter model. (D) Participant-level aggregation, penalized minus unpenalized selection, with unadjusted 95% paired mean-change intervals. Corrected signed-rank decisions are reported separately; a positive unadjusted mean interval is not a family-corrected penalty claim. Statistical families and differing model budgets are specified in Methods.

margin. The exploratory null-centred Holm analysis likewise supported no reduced count, even at the looser 0.03 margin. With only 10 participants and different hardware/tasks, it offers an uncertain boundary assessment rather than confirmation of the GRABMyo count. Derived bipolar pairs likewise failed to support a reduction from eight to seven at the primary margin. This representation cannot validate an equivalent reduction in physical bipolar sensors.

Full-array cross-session comparisons did not isolate placement displacement or reduced-layout robustness. Their participant results are retained in the Supplementary. A future independently collected same-montage cohort would be needed to validate a locked common layout, electrode count and online performance.

Table 3. Within-condition reduced minus full-array comparisons at the 0.02 margin. Null denotes the exploratory null-centred studentized bootstrap with Holm correction across tested counts. The RF experiment tests only 12 and 16 electrodes, so it estimates no minimum count.

Condition	n	k	$\bar{\Delta}$	95% CI	k^*_{CI}	k^*_{null}
Primary 250 ms	43	12	-0.0133	[-0.0160, -0.0106]	12	12
Rest included	43	12	-0.0139	[-0.0166, -0.0111]	12	12
150 ms	43	12	-0.0119	[-0.0145, -0.0093]	12	12
500 ms	43	12	-0.0126	[-0.0161, -0.0091]	12	12
NinaPro DB5	10	12	-0.0126	[-0.0200, -0.0056]	12	none
Derived bipolar	43	7	-0.0158	[-0.0203, -0.0113]	none	none
Random forest	43	12	-0.0078	[-0.0104, -0.0052]	not estimated	not estimated

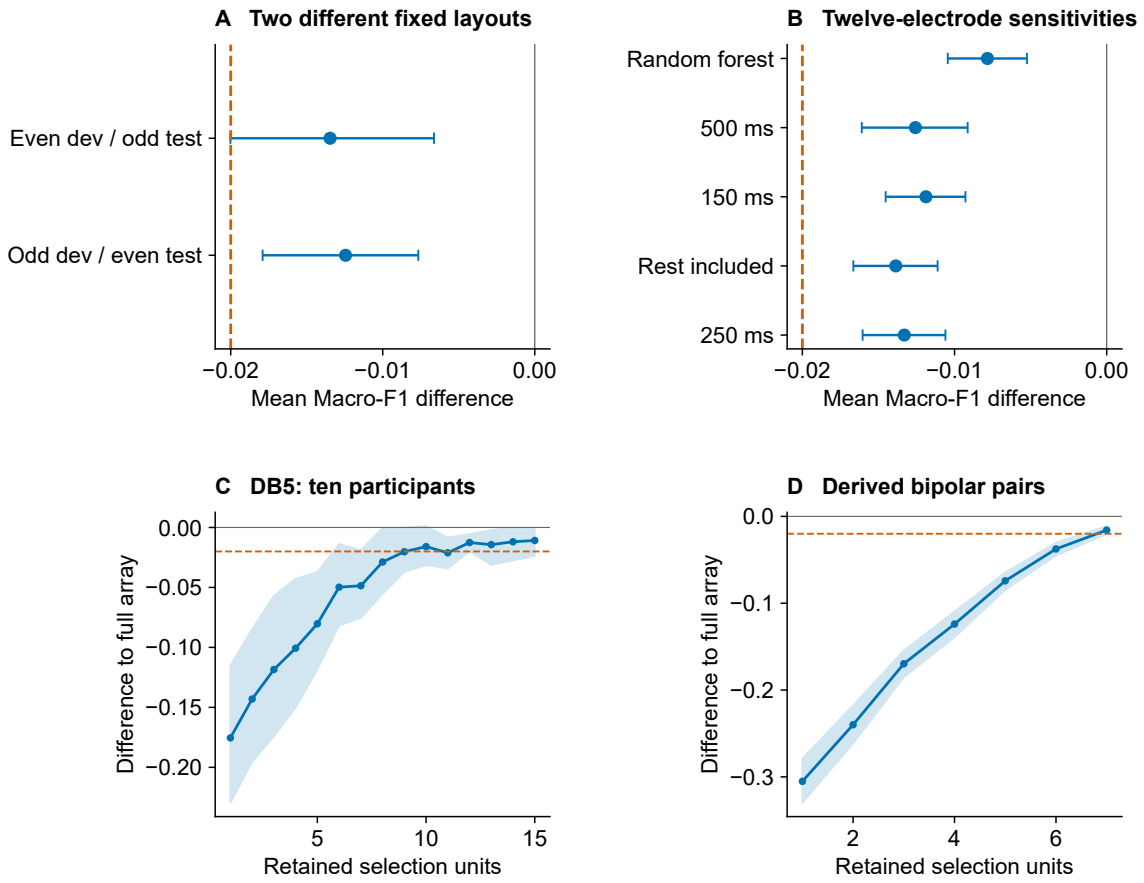


Figure 6. Boundaries of transfer. (A) Mean changes and intervals for two distinct split-derived layouts, using 21 and 22 test participants and all development windows for fitting. (B) Twelve-electrode changes for task, window and classifier sensitivities, with each condition's own sixteen-electrode reference. (C) DB5 reduction versus its sixteen-channel reference ($n = 10$). (D) Derived bipolar reduction versus its eight-pair reference ($n = 43$). The dashed line is the 0.02 margin. Every interval resamples participants; scores are not pooled across hardware or representations.

4 Discussion

The study supports removing one quarter of the GRABMyo monopolar recording sites within a defined mean recognition tolerance under this offline LOSO protocol. It does not show that every participant or gesture is preserved. The observed individual losses and corrected gesture declines make this distinction practically relevant: count selection for a real interface should include task priorities and user-specific failure criteria rather than relying solely on a pooled mean.

Attribution supplied predictive information beyond the mean random-subset baseline, and deletion with retraining supported its ranking relevance. Spatial coverage was a more competitive comparator. The matched primary experiment places geometric12's mean loss near the tolerance boundary; the direct mean advantage of SHAP12 remains uncertain. These results motivate attribution as one evaluated selection procedure, with additional computational cost, rather than establishing a universal advantage over simple geometry. The shared-model experiment found a positive SHAP mean advantage over grouped permutation at six retained electrodes, but no established mean advantage at twelve. The gain under more aggressive reduction does not mean that six electrodes met the primary tolerance. It changes permutation grouping, participant-disjoint validation, scaler isolation and resources together, so it cannot attribute historical differences solely to computational budget.

Stability regularization was not an established improvement. Five-fold pooling obscures some person heterogeneity, which motivated the participant-level extension, but more descriptive dispersion does not imply that subtracting it improves classification. All penalty settings were fixed and disclosed rather than selected on outer scores. Correlated electrode features, normalization and shared attributing models complicate physiological interpretation; high attribution is model reliance and can reflect redundancy, task discrimination or montage geometry without identifying a particular muscle.

The inference has two distinct limits. First, the tolerance is an engineering choice, not a clinically validated equivalence bound; tighter or looser tolerances support different counts. Second, participant bootstrap intervals are conditional on fitted models with strongly overlapping training sets. The null-centred sensitivity corrects how the bootstrap test distribution is constructed, but cannot remove training-set dependence or historical development exposure. A nested development cohort, independently locked tolerance and external validation would strengthen future inference.

Several aspects restrict transfer. Only able-bodied public datasets were analyzed, not target clinical users. Zero-phase filtering is non-causal. Complete rest/window curves retain the primary ranking; the classifier sensitivity also retains it and samples only two counts. DB5 uses different sensors, sampling, labels and a small cohort, and derived bipolar pairs are constructed signals. The two split layouts differ and count selection had already seen the cohort. Full-array cross-session results do not demonstrate reduced-layout displacement robustness. No real-time accuracy, latency, energy, hardware cost reduction or clinical benefit was measured.

A prospective follow-up should fix a single geometry and count using development data only, record new participants with the documented montage, and define participant- and gesture-level loss criteria before testing. It should evaluate causal preprocessing and a closed-loop task, recording prediction latency and electrode failures as measured outcomes. Such a protocol is a proposal, not evidence supplied by the present datasets.

5 Conclusion

Cross-fitted attribution supported reduction of the GRABMyo monopolar array to twelve electrodes at a 0.02 mean Macro-F1 tolerance under participant-independent offline evaluation. Utility beyond random selection was supported. Under shared resources, SHAP exceeded grouped permutation at six electrodes; its twelve-electrode mean advantage remained uncertain, as did the primary SHAP12–geometry12 contrast. Stability regularization did not establish an improvement. Participant losses, gesture-specific degradation, distinct fixed layouts and representation-dependent results restrict the claim to an evaluated reduction procedure. Independent validation is still required for a locked physical interface.

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Data availability
Public GRABMyo v1.1.0 recordings are available from PhysioNet (`https://doi.org/10.13026/89dm-f662`); NinaPro DB5 is available from `https://ninapro.hevs.ch`. The local revision package contains source code, protocols, participant results, predictions, numeric provenance and figure source data. [AUTHOR INPUT REQUIRED: permanent public archive URL/DOI before submission.]

Supplementary data
The separate Supplementary includes development departures, inference details, fixed-layout scope, ranking and redundancy diagnostics, complete machine-readable result tables and additional figures.

References
Yoshua Bengio and Yves Grandvalet. No unbiased estimator of the variance of k-fold cross-validation. *Journal of Machine Learning Research*, 5:1089–1105, 2004. URL `https://www.jmlr.org/papers/v5/grandvalet04a.html`.

A. C. Davison and D. V. Hinkley. *Bootstrap Methods and their Application*. Cambridge University Press, 1997. doi: 10.1017/cbo9780511802843.

Dario Farina, Ning Jiang, Hubertus Rehbaum, Ales Holobar, Bernhard Graimann, Hans Dietl, and Oskar C. Aszmann. The Extraction of Neural Information from the Surface EMG for the Control of Upper-Limb Prostheses: Emerging Avenues and Challenges. *IEEE Transactions on Neural Systems and Rehabilitation Engineering*, 22(4):797–809, 2014. doi: 10.1109/tnsre.2014.2305111.

Naveen Gehlot, Ashutosh Jena, Ankit Vijayvargiya, and Rajesh Kumar. Surface electromyography based explainable Artificial Intelligence fusion framework for feature selection of hand gesture recognition. *Engineering Applications of Artificial Intelligence*, 137:109119, 2024. doi: 10.1016/j.engappai.2024.109119.

Andrew Gelman and Hal Stern. The difference between significant and not significant is not itself statistically significant. *The American Statistician*, 60:328–331, 2006. doi: 10.1198/000313006X152649.

Ning Jiang, Ashirbad Pradhan, and Jiayuan He. Gesture Recognition and Biometrics ElectroMyogram (GRABMyo) (version 1.1.0). PhysioNet, 2024.

Scott Lundberg and Su-In Lee. A Unified Approach to Interpreting Model Predictions. In *Advances in Neural Information Processing Systems*, volume 30, 2017.

Scott M. Lundberg, Gabriel Erion, Hugh Chen, Alex DeGrave, Jordan M. Prutkin, Bala Nair, Ronit Katz, Jonathan Himmelfarb, Nisha Bansal, and Su-In Lee. From local explanations to global understanding with explainable AI for trees. *Nature Machine Intelligence*, 2(1):56–67, 2020. doi: 10.1038/s42256-019-0138-9.

Angkoon Phinyomark, Pornchai Phukpattaranont, and Chusak Limsakul. Feature reduction and selection for EMG signal classification. *Expert Systems with Applications*, 39(8):7420–7431, 2012. doi: 10.1016/j.eswa.2012.01.102.

Stefano Pizzolato, Luca Tagliapietra, Matteo Cognolato, Monica Reggiani, Henning Müller, and Manfredo Atzori. Comparison of six electromyography acquisition setups on hand movement classification tasks. *PLOS ONE*, 12(10):e0186132, 2017. doi: 10.1371/journal.pone.0186132.

Ashirbad Pradhan, Jiayuan He, and Ning Jiang. Multi-day dataset of forearm and wrist electromyogram for hand gesture recognition and biometrics. *Scientific Data*, 9(1), 2022. doi: 10.1038/s41597-022-01836-y.

Yali Qu, Haoyan Shang, Jing Li, and Shenghua Teng. Reduce Surface Electromyography Channels for Gesture Recognition by Multitask Sparse Representation and Minimum Redundancy Maximum Relevance. *Journal of Healthcare Engineering*, 2021:1–9, 2021. doi: 10.1155/2021/9929684.

Parul Rani, Sidharth Pancholi, Vikash Shaw, Suraj Pandey, Manfredo Atzori, and Sanjeev Kumar. Explainable AI-Guided Optimization of EMG Channels and Features for Precise Hand Gesture Classification: A SHAP-Based Study. *IEEE Transactions on Medical Robotics and Bionics*, 7(1):368–376, 2025. doi: 10.1109/tmrb.2024.3504007.

Erik Scheme and Kevin Englehart. Electromyogram pattern recognition for control of powered upper-limb prostheses: State of the art and challenges for clinical use. *The Journal of Rehabilitation Research and Development*, 48(6):643, 2011. doi: 10.1682/jrrd.2010.09.0177.

Donald J. Schuirmann. A comparison of the Two One-Sided Tests Procedure and the Power Approach for assessing the equivalence of average bioavailability. *Journal of Pharmacokinetics and Biopharmaceutics*, 15(6):657–680, 1987. doi: 10.1007/bf01068419.

Antonietta Stango, Francesco Negro, and Dario Farina. Spatial Correlation of High Density EMG Signals Provides Features Robust to Electrode Number and Shift in Pattern Recognition for Myocontrol. *IEEE Transactions on Neural Systems and Rehabilitation Engineering*, 23(2):189–198, 2015. doi: 10.1109/tnsre.2014.2366752.

Giorgio Visani, Enrico Bagli, Federico Chesani, Alessandro Poluzzi, and Davide Capuzzo. Statistical stability indices for LIME: Obtaining reliable explanations for machine learning models. *Journal of the Operational Research Society*, 73(1):91–101, 2022. doi: 10.1080/01605682.2020.1865846.

Ivan Vujaklija, Dario Farina, and Oskar Aszmann. New developments in prosthetic arm systems. *Orthopedic Research and Reviews*, Volume 8:31–39, 2016. doi: 10.2147/orr.s71468.

Esteban Walker and Amy S. Nowacki. Understanding Equivalence and Noninferiority Testing. *Journal of General Internal Medicine*, 26(2):192–196, 2011. doi: 10.1007/s11606-010-1513-8.

Feng Wang, Xiaohu Ao, Min Wu, Seiichi Kawata, and Jinhua She. Explainable deep learning for sEMG-based similar gesture recognition: A Shapley-value-based solution. *Information Sciences*, 672:120667, 2024. doi: 10.1016/j.ins.2024.120667.

Lanxin Xiang, Liang Shi, Youhui Ye, Boyu Jiang, Dawei Zhou, and Feng Guo. RoSHAP: A Distributional Framework and Robust Metric for Stable Feature Attribution. arXiv preprint arXiv:2605.15154, 2026.